CSE 270 Sports Analytics

PSS 1: Home advantage and power ratings

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Home advantage

Home advantage

In order to investigate the home advantage in nba, let's calculate

- Number of games the teams won and lost in home and away games separately.
- Average number of points scored by the teams in home and away games.
- Total number of points scored by the teams in home and away games.

```
nba_standings <- nba_east
data("nba2009_2021")

nba_won_count_h_t <- nba2009_2021 %>%
  group_by(home.TEAM_NAME, SEASON_ID) %>%
  summarize(H_L = sum(home.PTS < away.PTS),H_W = sum(home.PTS > away.PTS),
            MEAN_HP = mean(home.PTS), TOTAL_HP = sum(home.PTS)) %>%
  rename(TEAM_NAME = home.TEAM_NAME)

nba_won_count_a_t <- nba2009_2021 %>%
  group_by(away.TEAM_NAME, SEASON_ID) %>%
  summarize(A_L = sum(home.PTS > away.PTS),A_W = sum(home.PTS < away.PTS),
            MEAN_AP = mean(away.PTS), TOTAL_AP = sum(away.PTS)) %>%
  rename(TEAM_NAME = away.TEAM_NAME)
```

Home advantage

- Joining the two dataframes together to get the teams' statistics seperated in home and away games
- Joining the resulted dataframe with standings to get the teams rank that year.

```
nba_teams_stats <- inner_join(nba_won_count_h_t, nba_won_count_a_t,
                              by = c("TEAM_NAME", "SEASON_ID")) %>%
                              distinct()
nba_standings$Season <- as.character(nba_standings$Season)

nba_teams_stats <- inner_join(nba_teams_stats, nba_standings,
  by = c("TEAM_NAME" = "Team", "SEASON_ID" = "Season")) %>%
  arrange(SEASON_ID, Rank)

head(nba_teams_stats, 3)
```

```
## # A tibble: 3 x 17
## # Groups:   TEAM_NAME [3]
##   TEAM_N~1 SEASO~2   H_L   H_W MEAN_HP TOTAL~3   A_L   A_W MEAN_AP TOTAL~4  Rank
##   <chr>    <chr>   <int> <int>   <dbl>   <dbl> <int> <int>   <dbl>   <dbl> <int>
## 1 Clevela~ 2009        6    35    103.    4234    15    26   101.     4139     1
## 2 Boston ~ 2009       17    24    100.    4114    15    26    98.1    4022     2
## 3 Orlando~ 2009        7    34    105.    4322    16    25   100.     4104     3
## # ... with 6 more variables: W <int>, L <int>, Pct <dbl>, GB <chr>, PS <dbl>,
## #   PA <dbl>, and abbreviated variable names 1: TEAM_NAME, 2: SEASON_ID,
```

Home advantage

Let's define a few metrics for analysing the home advantage.

- On average home many more points are scored by the home teams **(MEAN_HP - MEAN_AP)**
- How many more games do the teams win in the games where they are the home team **(HW - A_W)**
- Percentage of games won in home games **(H_W / (H_W + A_W))**
- Percentage of points scored in home games **(TOTAL_HP / (TOTAL_HP + TOTAL_AP))**

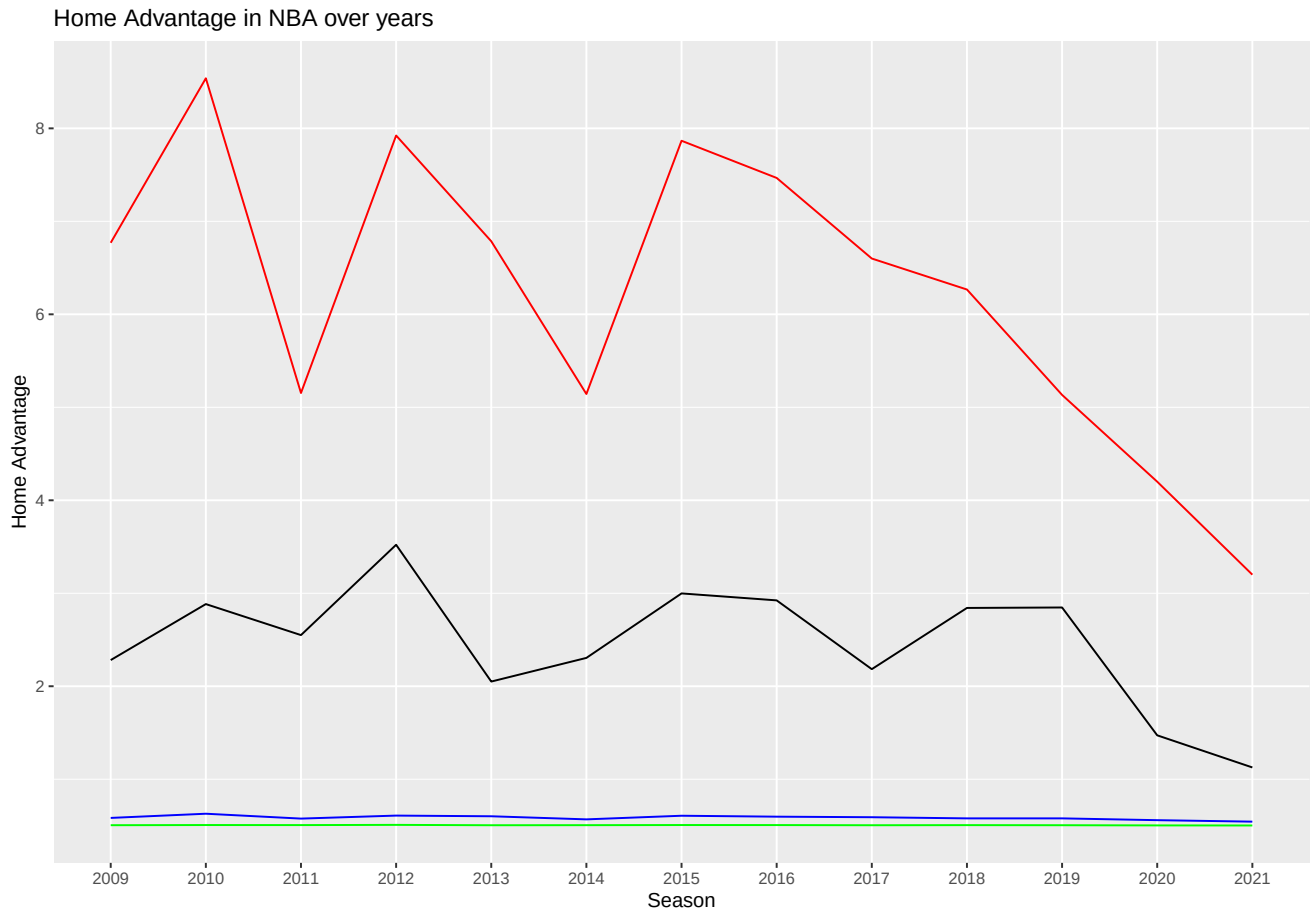
After calculating the metrics will count the averages and visualize over time.

```
nba_teams_stats_ha <- nba_teams_stats %>%
  mutate(
    HA1 = MEAN_HP - MEAN_AP,
    HA2 = H_W - A_W,
    HA3 = H_W /  (H_W + A_W) ,
    HA4 = TOTAL_HP / (TOTAL_HP + TOTAL_AP)) %>%
  rename_with(toupper)

ha_stats_by_season <- nba_teams_stats_ha %>%
  group_by(SEASON_ID) %>%
  summarise(mean_HA1 = mean(HA1), mean_HA2 = mean (HA2),
  mean_HA3 = mean(HA3), mean_HA4 = mean(HA4))
```

Home advantage

```
ggplot(ha_stats_by_season,aes(x = SEASON_ID, group = 1)) +
  geom_line(aes(y = mean_HA1)) + geom_line(aes(y = mean_HA2), color = "red") +
  geom_line(aes(y = mean_HA3), color = "blue") + geom_line(aes(y = mean_HA4), color = "green") +
  ggtitle("Home Advantage in NBA over years") + xlab('Season') + ylab('Home Advantage')
```



Home advantage

Interpretations from the previous plot.

- The ratios show almost the same result.
- The differences in between the number of points and number of games won varies over time and are independent of each other.

Scoring more points in home games does not guarantee to win more games at home. We can see multiple cases of decrease in the difference of the points that results to an increas in the difference of number of games won at home.

Let's try to pick the best metric for the home advantage by calculating the metric's correlation to the number of games won at home.

```
(cor(nba_teams_stats_ha$HA1, nba_teams_stats_ha$H_W))
```

```
## [1] 0.3520048
```

```
(cor(nba_teams_stats_ha$HA2, nba_teams_stats_ha$H_W))
```

```
## [1] 0.49641
```

```
(cor(nba_teams_stats_ha$HA3, nba_teams_stats_ha$H_W))
```

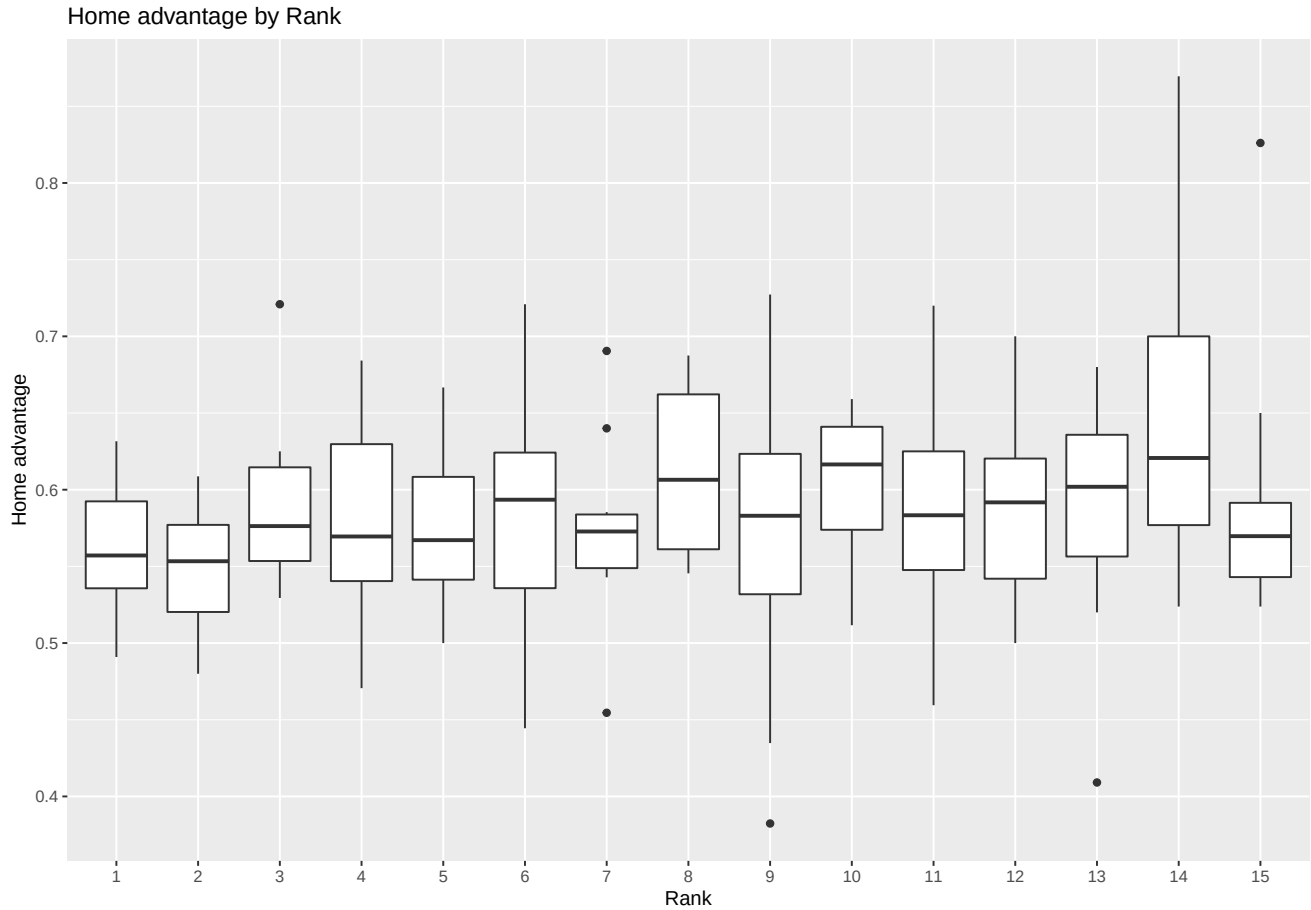
```
## [1] 0.0624086
```

```
(cor(nba_teams_stats_ha$HA4, nba_teams_stats_ha$H_W))
```

```
## [1] 0.09472873
```

Home advantage

```
ggplot(nba_teams_stats_ha, aes(x = as.factor(RANK), y = HA3)) +
  geom_boxplot() +
  ggtitle("Home advantage by Rank") +
  xlab("Rank") +
  ylab("Home advantage")
```



Home advantage

Interpretations from the previous plot.

- The home advantage gets balanced for the teams in the middle.
- The home advantage is not a key factor for teams in the top of the table.
- The teams in the bottom of the table less outliers,

Let's build a model to identify the significance of the rank for home advantage.

Home advantage

Home advantage is not a significant factor for the teams in the top of the table, but is a very important factor for the teams in the bottom.

```
nba_teams_stats_ha$RANK <- as.factor(nba_teams_stats_ha$RANK)

model <- lm(HA3 ~ RANK, data = nba_teams_stats_ha)

summary(model)
```

```
##
## Call:
## lm(formula = HA3 ~ RANK, data = nba_teams_stats_ha)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.18579 -0.04178 -0.00060  0.03778  0.23629
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.56285    0.01772  31.763  < 2e-16 ***
## RANK2       -0.01614    0.02506  -0.644  0.52024
## RANK3        0.02620    0.02506   1.045  0.29719
## RANK4        0.01831    0.02506   0.730  0.46603
## RANK5        0.01299    0.02506   0.518  0.60475
## RANK6        0.02168    0.02506   0.865  0.38811
## RANK7        0.01076    0.02608   0.413  0.68043
## RANK8        0.04709    0.02506   1.879  0.06179 .
## RANK9        0.00529    0.02506   0.211  0.83306
## RANK10       0.03962    0.02608   1.519  0.13046
## RANK11       0.01739    0.02554   0.681  0.49682
## RANK12       0.02810    0.02608   1.077  0.28279
## RANK13       0.02748    0.02506   1.096  0.27430
## RANK14       0.08106    0.02554   3.174  0.00176 **
## RANK15       0.02694    0.02608   1.033  0.30297
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Power Ratings

Power Ratings

Let's try to identify the relationship betwen power rating of the team and its chances of winning the game in National Hockey League.

```
nhl_games <- read.csv('../data/game.csv')

nhl_games <- nhl_games %>%
  mutate(outcome = ifelse(grepl("home win", outcome), "HW", "AW")) %>%
  select(game_id,season,date_time_GMT,home_team_id,away_team_id,
         home_goals, away_goals, outcome)

head(nhl_games,3)
```

```
##      game_id   season        date_time_GMT home_team_id away_team_id home_goals
## 1 2016020045 20162017 2016-10-19T00:30:00Z           16            4          7
## 2 2017020812 20172018 2018-02-07T00:00:00Z            7           24          3
## 3 2015020314 20152016 2015-11-24T01:00:00Z           52           21          1
##   away_goals outcome
## 1          4      HW
## 2          4      AW
## 3          4      AW
```

Power Ratings

Data preperation. Converging the dataset to long format to calculate cumulative goals scored and conceded.

```
nhl_games_long <- nhl_games %>%
  reshape2::melt(id.vars = c("game_id","season","date_time_GMT",
                             "home_team_id","away_team_id", "outcome"),
                 measure.vars = c("home_goals", "away_goals"),
                 variable.name = "h_a") %>%
  arrange(game_id)

nhl_games_long <- nhl_games_long %>%
  mutate(team1_id = ifelse(h_a == 'home_goals', home_team_id, away_team_id),
         team2_id = ifelse(h_a == 'home_goals', away_team_id, home_team_id)) %>%
  select(game_id,season,date_time_GMT,team1_id,team2_id,h_a, value,outcome)

head(nhl_games_long)
```

```
##      game_id   season        date_time_GMT team1_id team2_id        h_a value
## 1 2000020001 20002001 2000-10-04T23:00:00Z       25       21 home_goals     2
## 2 2000020001 20002001 2000-10-04T23:00:00Z       21       25 away_goals     2
## 3 2000020002 20002001 2000-10-05T23:00:00Z        6        9 home_goals     4
## 4 2000020002 20002001 2000-10-05T23:00:00Z        9        6 away_goals     4
## 5 2000020003 20002001 2000-10-05T23:00:00Z        7       16 home_goals     4
## 6 2000020003 20002001 2000-10-05T23:00:00Z       16        7 away_goals     2
```

Power Ratings

Calculating the sum of the goals scored and conceded by each team in the last 5 games before each game.

```
nhl_games_long <- nhl_games_long %>%
  group_by(season, team1_id) %>%
  mutate(cum_sc = rollsum(value, k = 5, fill = NA, align="right")) %>%
  ungroup() %>%
  group_by(season, team2_id) %>%
  mutate(cum_c = rollsum(value, k = 5, fill = NA, align="right")) %>%
  tidyr::drop_na() %>%
  as.data.frame()

head(nhl_games_long,3)
```

```
##      game_id   season        date_time_GMT team1_id team2_id        h_a value
## 1 2000020052 20002001 2000-10-13T23:30:00Z       26        6 home_goals     5
## 2 2000020052 20002001 2000-10-13T23:30:00Z        6       26 away_goals     0
## 3 2000020058 20002001 2000-10-14T23:00:00Z       25       15 home_goals     3
##   outcome cum_sc cum_c
## 1      HW     23    14
## 2      HW     16    11
## 3      HW     13    16
```

Power Ratings

Converting back to wide format and calculating the power ratings.

- Home team's power rating = Number of goals scored in the last 5 games - Number of goals conceded in the last 5 games.
- The same calculation is done for away team.
- Calculating the power difference of the game.

```
nhl_games_wide = reshape(nhl_games_long, direction = "wide", timevar = "h_a",
  idvar = "game_id", v.names = c("value","cum_sc", "cum_c")) %>%
  mutate(powerhome = cum_sc.home_goals - cum_c.home_goals,
         poweraway = cum_sc.away_goals - cum_c.away_goals,
         powerdiff = powerhome - poweraway) %>%
  select(game_id,season,date_time_GMT,team1_id,team2_id, value.home_goals,
       value.away_goals,cum_sc.home_goals, cum_c.home_goals,
       cum_sc.away_goals,cum_c.away_goals, powerhome, poweraway, powerdiff, outcome) %>%
  tidyr::drop_na()

head(nhl_games_wide, 3)
```

```
##      game_id   season        date_time_GMT team1_id team2_id value.home_goals
## 1 2000020052 20002001 2000-10-13T23:30:00Z       26        6                5
## 2 2000020058 20002001 2000-10-14T23:00:00Z       25       15                3
## 3 2000020065 20002001 2000-10-15T23:00:00Z       17       20                2
##   value.away_goals cum_sc.home_goals cum_c.home_goals cum_sc.away_goals
## 1                0                23               14                16
## 2                0                13               16                 8
## 3                4                14               12                12
##   cum_c.away_goals powerhome poweraway powerdiff outcome
## 1               11         9         5         4      HW
## 2               10        -3        -2        -1      HW
## 3               13         2        -1         3      AW
```

Power Ratings

Calculating the frequencies

```
freqs = dcast(nhl_games_wide, powerdiff ~ outcome)
head(freqs, 3)
```

```
##   powerdiff AW HW
## 1       -36  1  0
## 2       -32  0  1
## 3       -30  0  1
```

Power Ratings

Calculating the relative frequencies

```
freqs_rel = data.frame(freqs, freqs[,2:3]/rowSums(freqs[,2:3]))
head(freqs_rel)
```

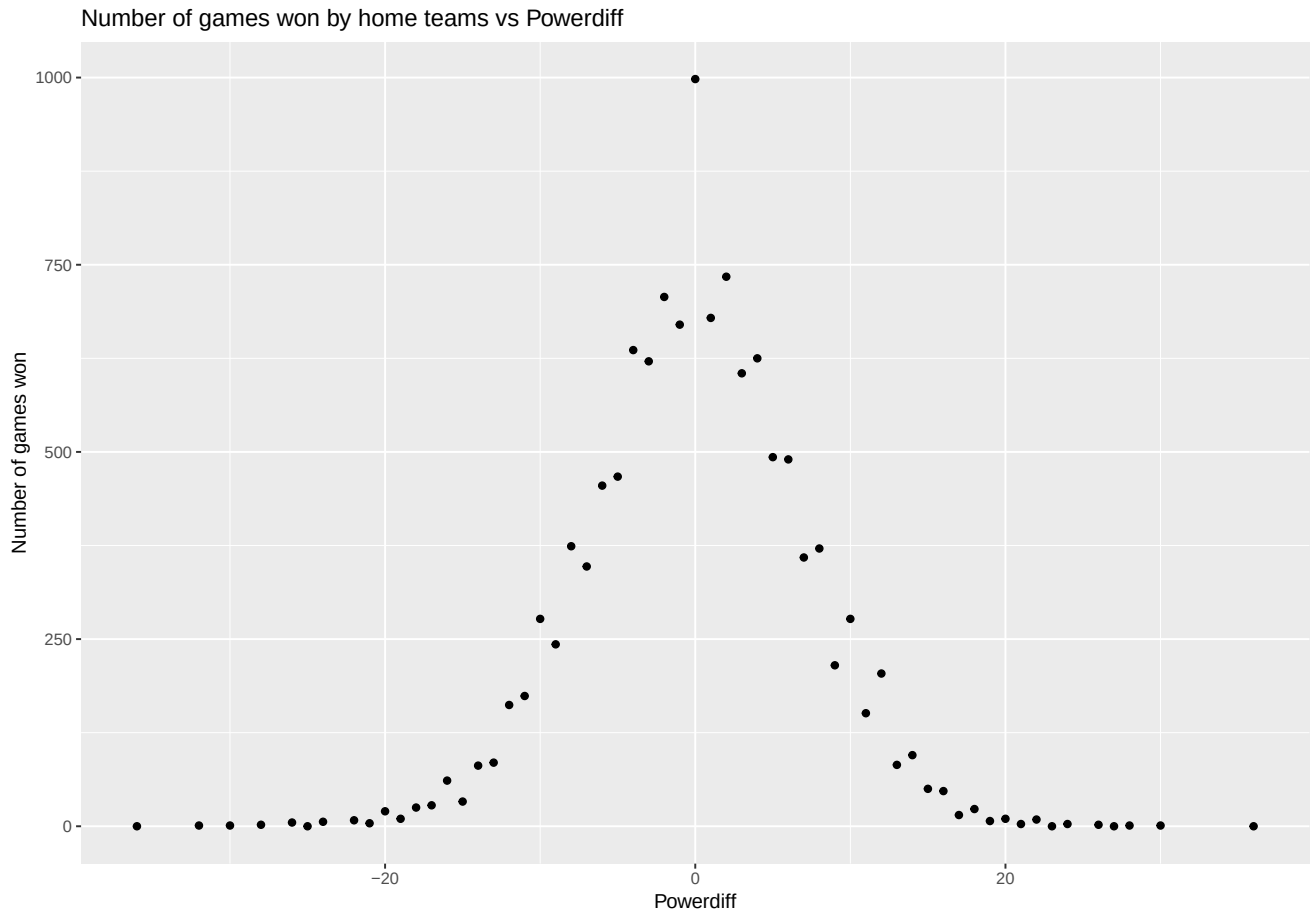
```
##   powerdiff AW HW      AW.1      HW.1
## 1       -36  1  0 1.0000000 0.0000000
## 2       -32  0  1 0.0000000 1.0000000
## 3       -30  0  1 0.0000000 1.0000000
## 4       -28  2  2 0.5000000 0.5000000
## 5       -26  2  5 0.2857143 0.7142857
## 6       -25  2  0 1.0000000 0.0000000
```

```
freqs_rel[freqs_rel$powerdiff == 0,]
```

```
##    powerdiff  AW  HW      AW.1      HW.1
## 30         0 855 998 0.4614139 0.5385861
```

Power Ratings

```
ggplot(freqs_rel, aes(x = powerdiff, y = HW)) +
  geom_point() +
  ggtitle("Number of games won by home teams vs Powerdiff") +
  xlab("Powerdiff") +
  ylab("Number of games won")
```



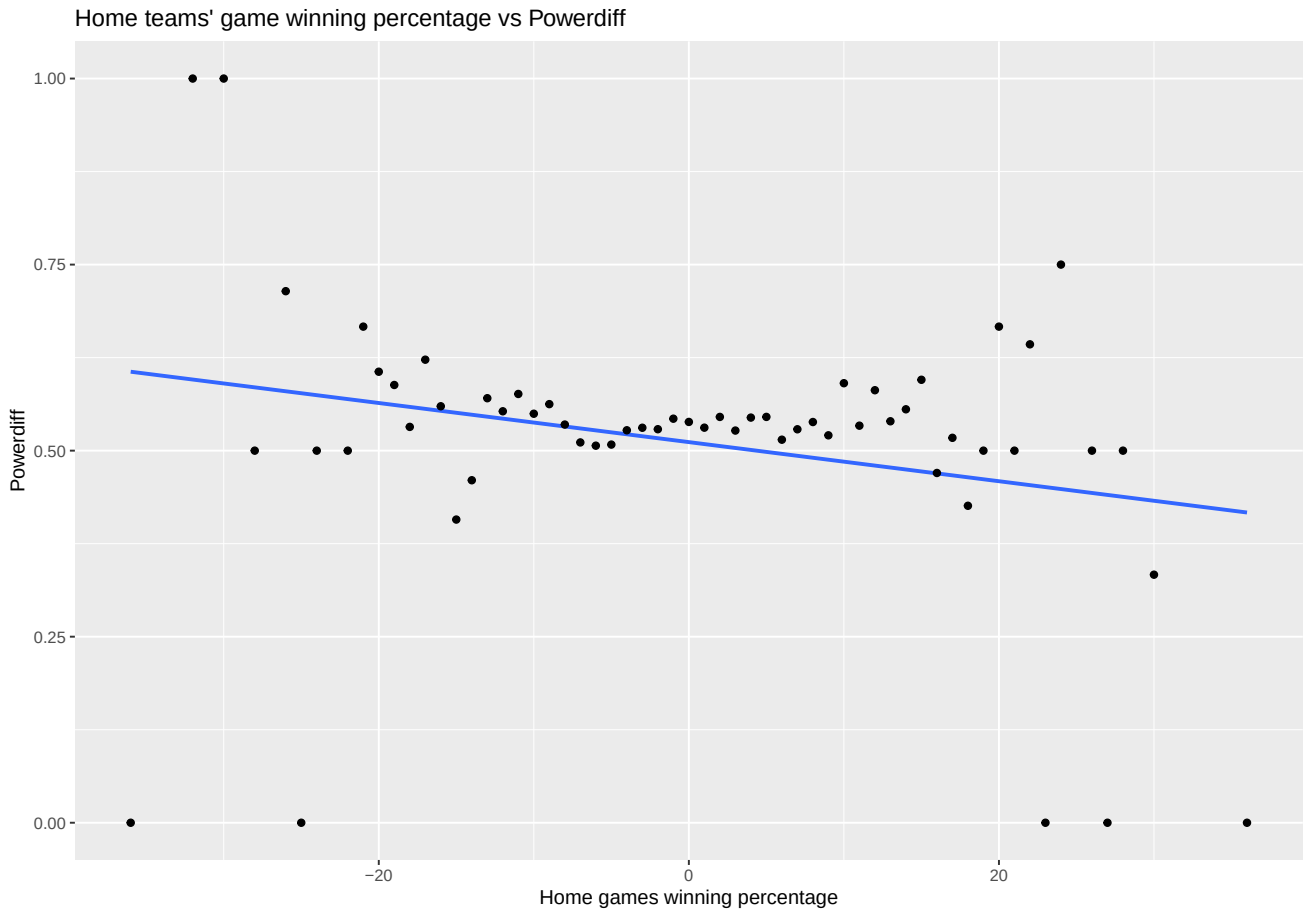
Power Ratings

Interpretations from the previous plot.

- With the increase of power rating the percentage of the games won by the home team increases.
- Very few games when the powerdiff was a high number in favor of the home or away team.

Power Ratings

```
ggplot(freqs_rel, aes(x = powerdiff, y = HW.1)) +
  geom_smooth(method = "lm", se = F) +
  geom_point() +
  ggtitle("Home teams' game winning percentage vs Powerdiff") +
  xlab("Home games winning percentage") +
  ylab("Powerdiff")
```



Power Ratings

Interpretations from the previous plot.

- The variables do not seem to have a relationship.

Power Ratings

- Small correlation coefficient.
- Almost no significance for the powerdiff and weak predictive power.

```
(cor(freqs_rel$powerdiff, freqs_rel$HW.1))
```

```
## [1] -0.2510591
```

```
summary(lm(HW.1 ~ powerdiff, data = freqs_rel))
```

```
##
## Call:
## lm(formula = HW.1 ~ powerdiff, data = freqs_rel)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.60606 -0.01763  0.02712  0.05393  0.40971
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.511465   0.024056  21.261   <2e-16 ***
## powerdiff   -0.002628   0.001342  -1.958   0.0551 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1848 on 57 degrees of freedom
## Multiple R-squared:  0.06303,    Adjusted R-squared:  0.04659
## F-statistic: 3.834 on 1 and 57 DF,  p-value: 0.05511
```

Power Ratings

Adjusting the calculation strategy to have a power rating relative to the opponent.

- Home team's power rating = Number of goals scored in the last 5 games - Number of goals conceded in the last 5 games by the opponent.
- The same calculation is done for away team.
- Calculating the power difference of the game.

```
nhl_games_wide = reshape(nhl_games_long, direction = "wide", timevar = "h_a",
  idvar = "game_id", v.names = c("value","cum_sc", "cum_c")) %>%
  mutate(powerhome = cum_sc.home_goals - cum_c.away_goals,
         poweraway = cum_sc.away_goals - cum_c.home_goals,
         powerdiff = powerhome - poweraway) %>%
  select(game_id,season,date_time_GMT,team1_id,team2_id, value.home_goals,
       value.away_goals,cum_sc.home_goals, cum_c.home_goals,
       cum_sc.away_goals,cum_c.away_goals, powerhome, poweraway, powerdiff, outcome) %>%
  tidyr::drop_na()

head(nhl_games_wide, 3)
```

```
##      game_id   season        date_time_GMT team1_id team2_id value.home_goals
## 1 2000020052 20002001 2000-10-13T23:30:00Z       26        6                5
## 2 2000020058 20002001 2000-10-14T23:00:00Z       25       15                3
## 3 2000020065 20002001 2000-10-15T23:00:00Z       17       20                2
##   value.away_goals cum_sc.home_goals cum_c.home_goals cum_sc.away_goals
## 1                0                23               14                16
## 2                0                13               16                 8
## 3                4                14               12                12
##   cum_c.away_goals powerhome poweraway powerdiff outcome
## 1               11        12         2        10      HW
## 2               10         3        -8        11      HW
## 3               13         1         0         1      AW
```

Power Ratings

Calculating the relative frequencies

```
freqs = dcast(nhl_games_wide, powerdiff ~ outcome)
freqs_rel = data.frame(freqs, freqs[,2:3]/rowSums(freqs[,2:3]))
head(freqs_rel)
```

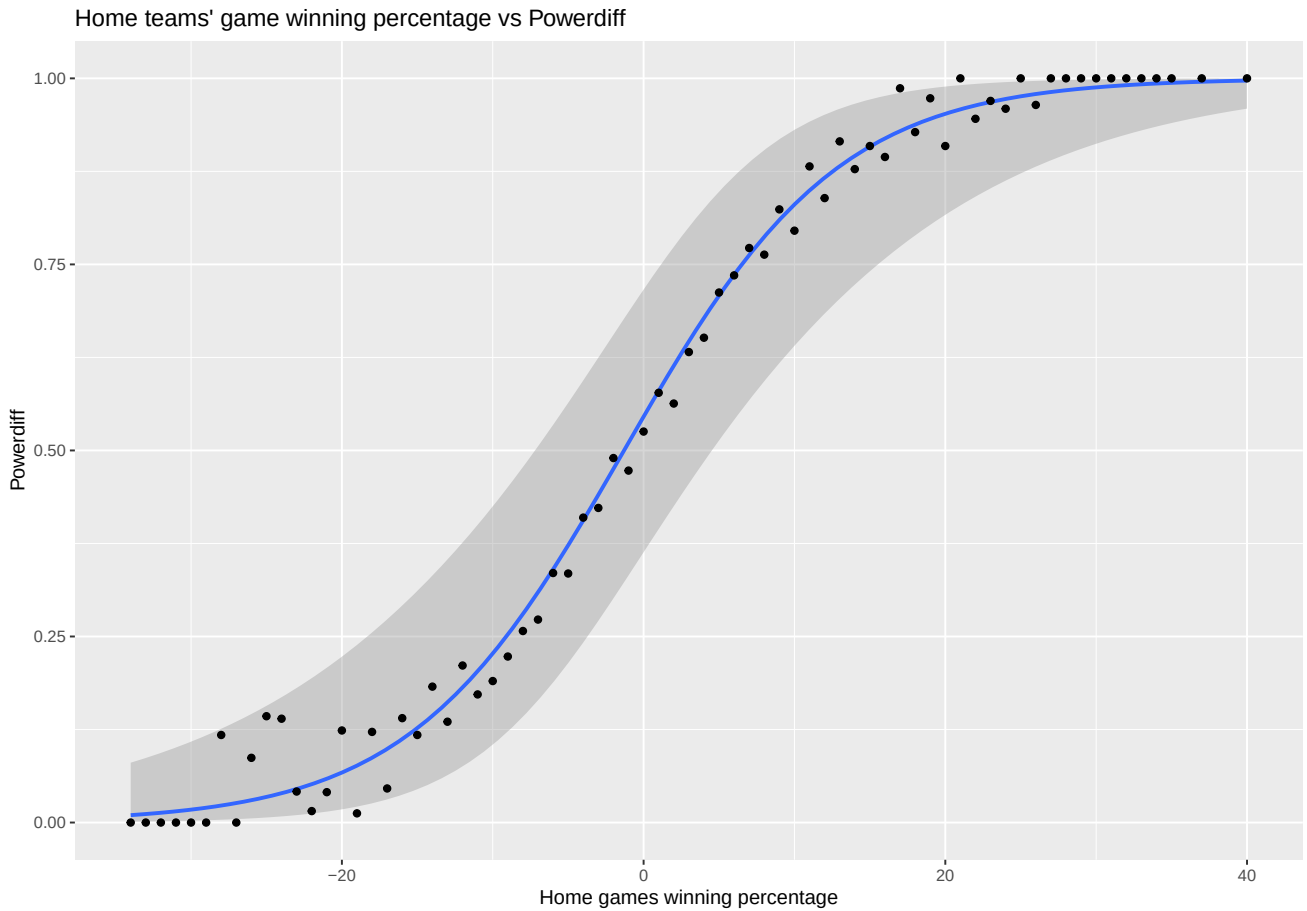
```
##   powerdiff AW HW AW.1 HW.1
## 1       -34  1  0    1    0
## 2       -33  1  0    1    0
## 3       -32  8  0    1    0
## 4       -31  1  0    1    0
## 5       -30  7  0    1    0
## 6       -29  3  0    1    0
```

```
freqs_rel[freqs_rel$powerdiff == 0,]
```

```
##    powerdiff  AW  HW      AW.1      HW.1
## 35         0 533 590 0.4746215 0.5253785
```

Power Ratings

```
ggplot(freqs_rel, aes(x = powerdiff, y = HW.1)) +
  geom_smooth(method = "glm",
    method.args = list(family = "binomial"),
    se = TRUE) +
  geom_point() +
  ggtitle("Home teams' game winning percentage vs Powerdiff") +
  xlab("Home games winning percentage") +
  ylab("Powerdiff")
```



Power Ratings

Interpretations from the previous plot.

- The variables are now strongly related.
- The relationship is not linear.
- Logistic regression model fits the data very well.

Let's try fitting a linear model.

```
(cor(freqs_rel$powerdiff, freqs_rel$HW.1))
```

```
## [1] 0.9672485
```

```
summary(lm(HW.1 ~ powerdiff, data = freqs_rel))
```

```
##
## Call:
## lm(formula = HW.1 ~ powerdiff, data = freqs_rel)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.23281 -0.08317  0.01437  0.08750  0.17274
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.5102467  0.0118633   43.01   <2e-16 ***
## powerdiff   0.0180641  0.0005666   31.88   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1004 on 70 degrees of freedom
## Multiple R-squared:  0.9356, Adjusted R-squared:  0.9346
## F-statistic:  1016 on 1 and 70 DF,  p-value: < 2.2e-16
```

Power Ratings

Fitting a logarithmic regession (logit) model.

- Strong predictive power.
- High significance for powerdiff.

Better to take the linear model, as the difference is not huge and linear also performs pretty good and in addition it is easier to interpret.

```
model <- glm(HW.1 ~ powerdiff, data = freqs_rel, family = binomial(link = "logit"))
(r_squared <- with(summary(model), 1 - deviance/null.deviance))
```

```
## [1] 0.9687526
```

```
(summary(model))
```

```
##
## Call:
## glm(formula = HW.1 ~ powerdiff, family = binomial(link = "logit"),
##     data = freqs_rel)
##
## Deviance Residuals:
##      Min        1Q    Median        3Q       Max
## -0.29596  -0.08045  -0.01349   0.11166   0.45370
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.18216    0.37938   0.480    0.631
## powerdiff    0.14070    0.03157   4.456 8.34e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 55.7252  on 71  degrees of freedom
```